Andrew Brettin

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Climate data scientist specializing in applying probabilistic machine learning to forecast and quantify sea level risks. Expertise in scalable data pipelines, uncertainty quantification, extreme value analysis, forecasting, and visualization.

Experience

Courant Institute of Mathematical Sciences, New York University — New York, NY
Postdoctoral Research Associate — *May 2025–September 2025*
Graduate Research Assistant — *September 2019–May 2025*
Topics: Machine learning for quantifying climate risks and sea level predictability.

Mathematics and Climate Research Network, University of Minnesota — Minneapolis, MN
Undergraduate Researcher — *September 2017–May 2019*
Topics: Mathematical modeling of biodiversity in ecological systems.

Center for Discrete Mathematics and Computer Science, Rutgers University — New Brunswick, NJ
NSF Undergraduate Research Intern — *May 2018–July 2018*
Topics: Stochastic models of cancer growth. Reaction-diffusion equations, Monte Carlo simulation.

University of North Carolina, Greensboro — Greensboro, NC
NSF Undergraduate Research Intern — *May 2017–July 2017*
Topics: Epidemiological models of disease transmission. Differential equations, game theory, sensitivity analysis.

Education

Ph.D., Courant Institute, NYU. Atmosphere-Ocean Science & Mathematics. Advisor: Laure Zanna — *May 2025*

B.Sc., University of Minnesota. Mathematics, *summa cum laude with high distinction (GPA 3.92)* — *May 2019*

Awards and distinctions: VoLo Fellowship (2020) • Henry M. MacCracken Fellowship (2019) • Hans H. Dalaker Scholarship (2018) • Gold Scholar Award (2015)

Research Projects

Estimating temperature and precipitation extremes with quantile neural networks — *2025*

- Designed and benchmarked a novel machine learning framework for uncertainty quantification, outperforming industry-standard baselines across 1,500+ weather stations.
- Validated model robustness using probabilistic evaluation metrics (CRPS, calibration), minimizing forecast bias and improving confidence interval reliability.

Outcomes: [preprint] [code]

Learning efficient forecasts for regional sea surface height dynamics — *2024–2025*

- Customized a deep learning framework (convolutional autoencoders) with nonlinear dynamical systems theory (Koopman operators) to transform high-dimensional, complex spatiotemporal data into a tractable, linear latent representation.
- Implemented Distributed Data Parallelism (DDP) in Pytorch to efficiently scale training over 200+ GB of spatiotemporal data on multiple GPUs.
- Demonstrated superior predictive performance over stochastic Linear Inverse Models (LIM), improving forecasting skill by 5–10% using latent representations of similar complexity.

Outcomes: [Article, *Geophysical Research Letters*] [code] [presentation]

Identifying sources of sea level predictability with mean-variance networks — *2023–2024*

- Architected scalable ETL pipelines to process TB-scale spatiotemporal climate datasets, leveraging Dask and Xarray for parallel processing on high-performance computing (HPC) systems
- Automated end-to-end training and optimization for an ensemble of 6,500+ neural networks.
- Identified physical drivers of sea level predictability using Explainable AI (XAI) techniques

Outcomes: [Article, *Artificial Intelligence for the Earth Systems*] [code] [conference presentation]

Exploring nonstationarity of coastal sea level distributions *2021–2022*

- Developed a non-stationary statistical framework for quantifying shifting probability distributions in tide gauge observations, enabling more robust risk assessment of extreme events under shifting climate states.

Outcomes: [Article, *Environmental Data Science*] [code] [presentation]

Technical Skills

Languages	Python • Bash • Julia • R • MATLAB • C++ • LaTeX
MLOps	git/GitHub • Cursor • Singularity • Pytorch/TensorFlow/Keras • Lightning AI • Weights & Biases
Computation	HPC • Dask/xarray • Distributed computing (SLURM/PBS/GCP) • Data parallelism
Stats/ML	Linear/logistic regression • Extreme value analysis • Quantile regression • PCA • Unsupervised learning • Gaussian processes • Random forests • Statistical dynamical models (LIM, EnKF, DMD) • Numerical methods (optimization, interpolation, finite difference, spectral methods, MCMC)
Deep learning	CNNs • PINNs • Autoencoders • U-nets • transformers
Visualization	matplotlib • seaborn • cartopy • HoloViews • Plotly Dash

Publications

1. **Brettin, A.** *Data-Driven Approaches for Predictability and Uncertainty Quantification of Sea Level Variability.* 3223742845. PhD thesis (New York University, 2025).
2. **Brettin, A.**, Zanna, L. & Barnes, E. A. Learning Propagators for Sea Surface Height Forecasts Using Koopman Autoencoders. *Geophysical Research Letters* **52.** 10.1029/2024GL112835, e2024GL112835 (2025).
3. **Brettin, A.**, Zanna, L. & Barnes, E. A. Uncertainty-Permitting Machine Learning Reveals Sources of Dynamic Sea Level Predictability across Daily to Seasonal Time Scales. *Artificial Intelligence for the Earth Systems* **4.** 10.1175/AIES-D-25-0014.1, 250014 (2025).
4. Falasca, F., **Brettin, A.**, Zanna, L., Griffies, S. M., Yin, J. & Zhao, M. Exploring the nonstationarity of coastal sea level probability distributions. *Environmental Data Science* **2.** 10.1017/eds.2023.10, e16 (2023).
5. Meyer, K., Broda, J., **Brettin, A.**, Sánchez Muñiz, M., Gorman, S., Isbell, F., Hobbie, S. E., Zeeman, M. L. & McGehee, R. Nitrogen-induced hysteresis in grassland biodiversity: a theoretical test of litter-mediated mechanisms. *The American Naturalist* **201.** 10.1086/724383, E153–E167 (2023).
6. **Brettin, A.**, Rossi-Goldthorpe, R., Weishaar, K. & Erovenko, I. V. Ebola could be eradicated through voluntary vaccination. *Royal Society Open Science* **5.** 10.1098/rsos.171591, 171591 (2018).

Additional Activities

- **NASA/JPL Summer School on Satellite Observations and Climate Models** *2023*
  Keck Institute for Space Studies, Caltech, Pasadena, CA
- **LEAP Momentum Bootcamp on Climate Data Science** *2022*
  Columbia University, New York, NY
- **Volunteer tutor, math grades 5–8** *2021–2022*
  Common Denominator, New York, NY
- **Vice President, Courant Student Organization** *2021–2022*
  Courant Institute of Mathematical Sciences, New York University, New York, NY
- **Project mentor, Undergraduate Research Program in Data Science** *2021*
  NYU Center for Data Science (collaboration with the National Society of Black Physicists), New York, NY
- **Reviewer**, *Geophysical Research Letters* (2025–) and *Journal of Geophysical Research: Machine Learning and Computation* (2025–)